10/16/2026: Recurrence

CSCI 246: Discrete Structures

Textbook reference: Sec 23, Scheinerman

Quiz Retake Pickup

Quiz retakes are in the front of the room, grouped into four bins (A-G, H-L, M-R, S-Z) by last name. The midterms are upside down with your last name on the back. Come find yours before, during, or after class. Only turn the quiz over if it's yours.

Announcements

1. If you answered the playlists problem on the midterm in terms of combinations rather than permutations, see me.
2. Detail on induction step: we assume any k **greater than or equal to the base**, and then show the claim must hold for $k + 1$.

Today's Agenda

- Weekly quiz (20 mins)
- Mini lecture ($\approx$ 15 mins)
- Group exercises ($\approx$ 15 mins)

Weekly Quiz #6

1. *(Sec. 17 – Binomial Coefficients)* Evaluate $\binom{5}{0}$, $\binom{5}{1}$ and $\binom{5}{2}$.
2. *(Sec. 20 – Contradiction and contraposition.)* Prove by contraposition that if n^2 is even, then n is even.
3. *(Extra credit opportunity: You will get 10% extra credit on your midterm if you can answer this proof question perfectly.)* Let x be an integer. Prove that $0 \mid x$ if and only if $x = 0$.

Reference Material: Definition

Let $n, k \in \mathbb{N}$. The symbol $\binom{n}{k}$ denotes the number of k-element subsets of an n-element set.

Mini-lecture on recurrence

Tower of Hanoi

In 1883, French mathematician Édouard Lucas invented a puzzle, based on an old Indian legend, called the **Tower of Hanoi**.



Prize. The puzzle offered 10,000 francs (about $45,000 USD today) to anyone who could solve the puzzle with 64 disks.

Tower of Hanoi

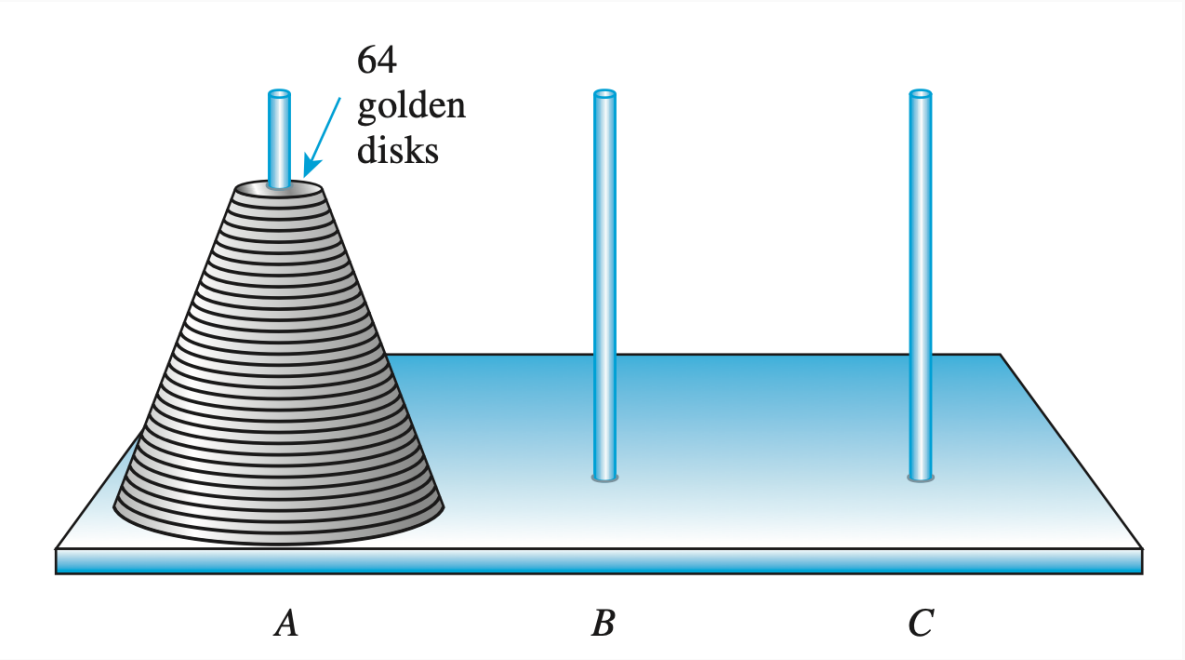
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Legend. According to the legend, when the puzzle is solved, the world will end.

Tower of Hanoi

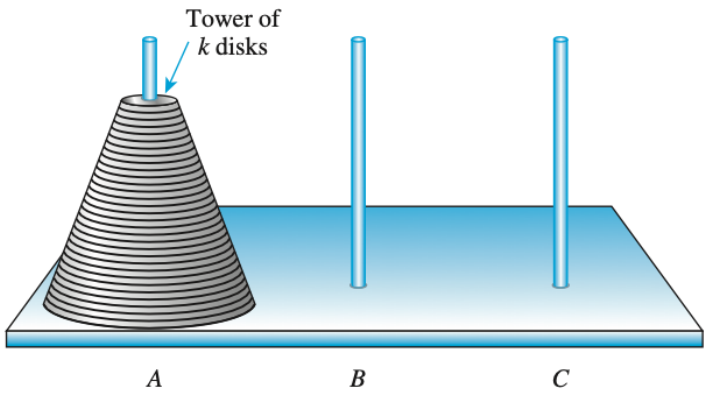


Rules.

- You must move all disks one by one from one pole to another.
- You must never place a larger disk on top of a smaller one.

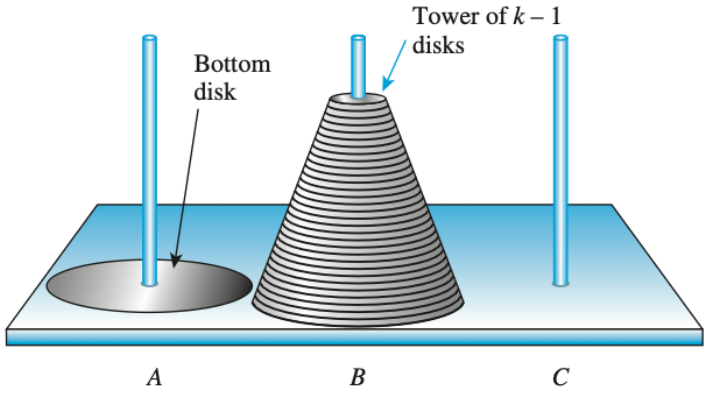
Tower of Hanoi: Solution As A Recursion

Idea. Suppose you know the optimal solution for $k-1$ disks. Then here's the optimal solution for k disks.



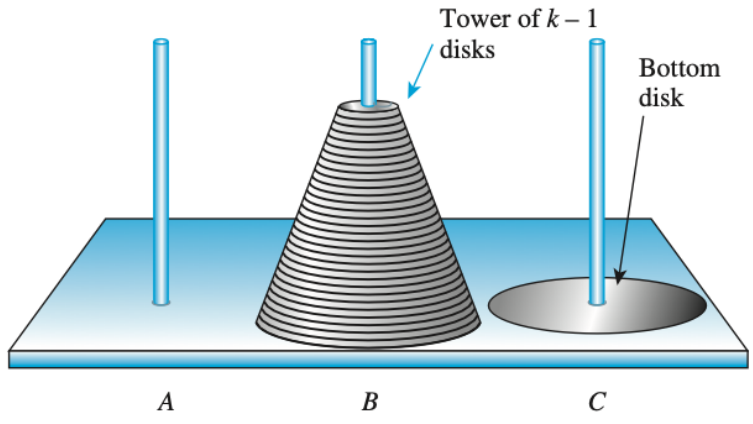
Initial Position

(a)



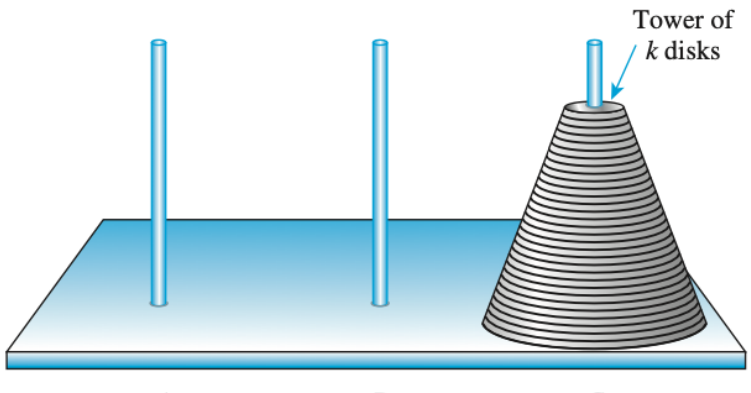
Position after Transferring $k-1$ Disks from A to B

(b)



Position after Moving the Bottom Disk from A to C

(c)



Position after Transferring $k-1$ Disks from B to C

(d)

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Solution. From the last slide, we have

$$\begin{aligned} m_k &= m_{k-1} + 1 + m_{k-1} \\ &= 2m_{k-1} + 1 \end{aligned}$$

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(Note: This solution can be verified by induction!)

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Poll. What does this say about the prospects for winning the prize?

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Using this, we find $m_{64} \approx 1.88 \times 10^{19}$. In other words, if you work rapidly, and move one disk every second, the total number of time it would take to solve the puzzle with 64 disks is: 584.5 billion years!

Recursions are everywhere!

Some examples from textbooks:

- Fibonacci sequence (e.g., population growth)
- Compound interest





Some examples from real-life software:

- Reinforcement learning (e.g., computer chess)
- Early cancer detection





Recursive programs are slow

Consider the program to solve $a_n = 3a_{n-1} - 2a_{n-2}$ with initial conditions $a_0 = 1, a_1 = 5$:

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Problem. This program is expensive to run! To get the 50-th term a_{50}, the program must be called $b_{50} \approx 40.7$ billion times!


Is there a better way
to handle recursions?

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Evaluation.  This is the best situation to be in. Clear and quick. The textbook gives techniques for obtaining these.

Most important results from the textbook – Part 1

Solving first-order recurrences (Props 23.1, 23.3)

All solutions to recurrence relation $a_n = sa_{n-1} + t$ have the form

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Solving second-order recurrences (Thms 23.5, 23.9)

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$$a_n = \begin{cases} c_1 r_1^n + c_2 r_2^n, & \text{if } r_1 \neq r_2 \\ c_1 r^n + c_2 n r^n, & \text{if } r_1 = r_2 := r \end{cases}$$

where r_1, r_2 are roots of $x^2 - s_1 x - s_2 = 0$.

Most important results from the textbook – Part 2

Thm 23.17 shows how to find a polynomial expression to generate the first few terms of a sequence.

Theorem 23.17 Let $a_0, a_1, a_2, \ldots$ be a sequence of numbers. The terms a_n can be expressed as polynomial expressions in n if and only if there is a nonnegative integer k such that for all $n \geq 0$ we have $\Delta^{k+1} a_n = 0$. In this case,

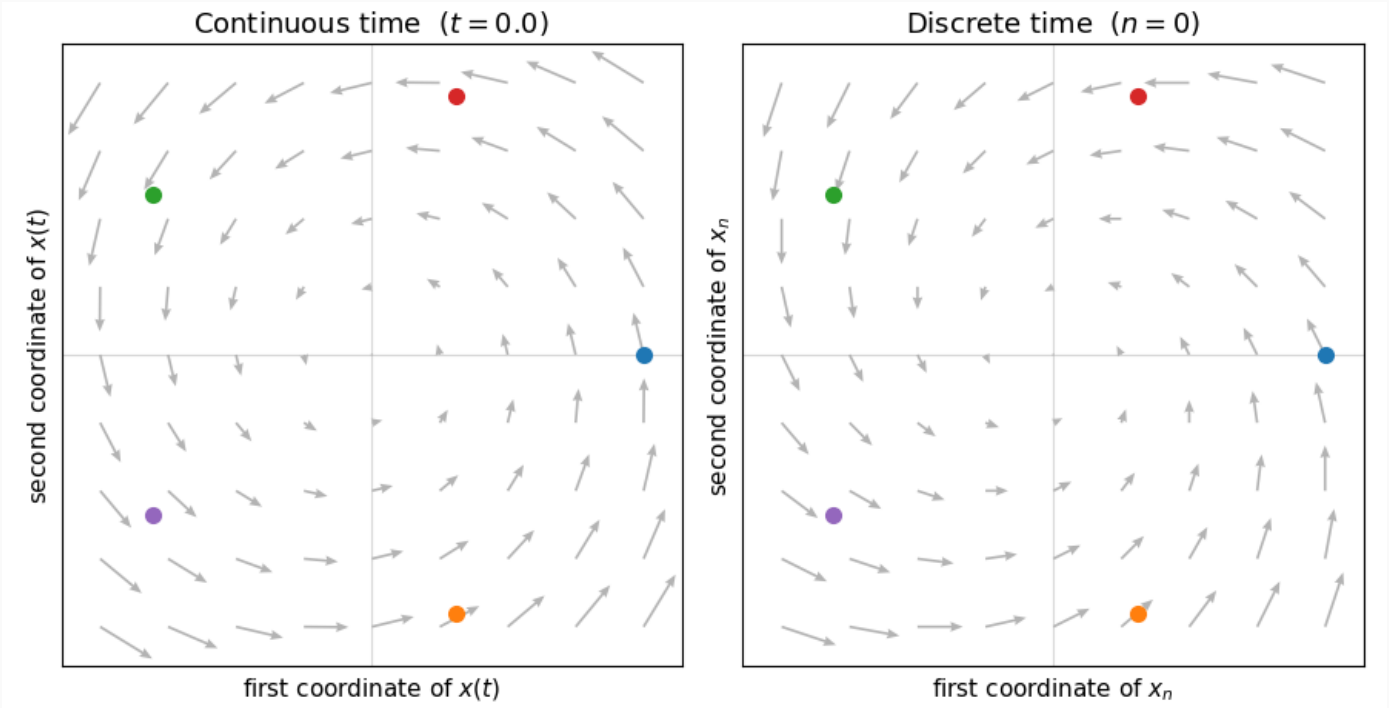
$$a_n = a_0 \binom{n}{0} + (\Delta a_0) \binom{n}{1} + (\Delta^2 a_0) \binom{n}{2} + \cdots + \left(\Delta^k a_0\right) \binom{n}{k}.$$

Fun Note: Recurrences are discrete-time differential equations

Continuous time (differential eq.)
$x'(t) = k\,x(t) \quad \Rightarrow \quad x(t) = x(0)\,e^{kt}$

Discrete time (recurrence)
$a_n = s\,a_{n-1} \quad \Rightarrow \quad a_n = a_0\,s^n$

Below, each coordinate obeys $a_n = 1.6\,a_{n-1} - 0.8\,a_{n-2}$, which gives **spirals**.

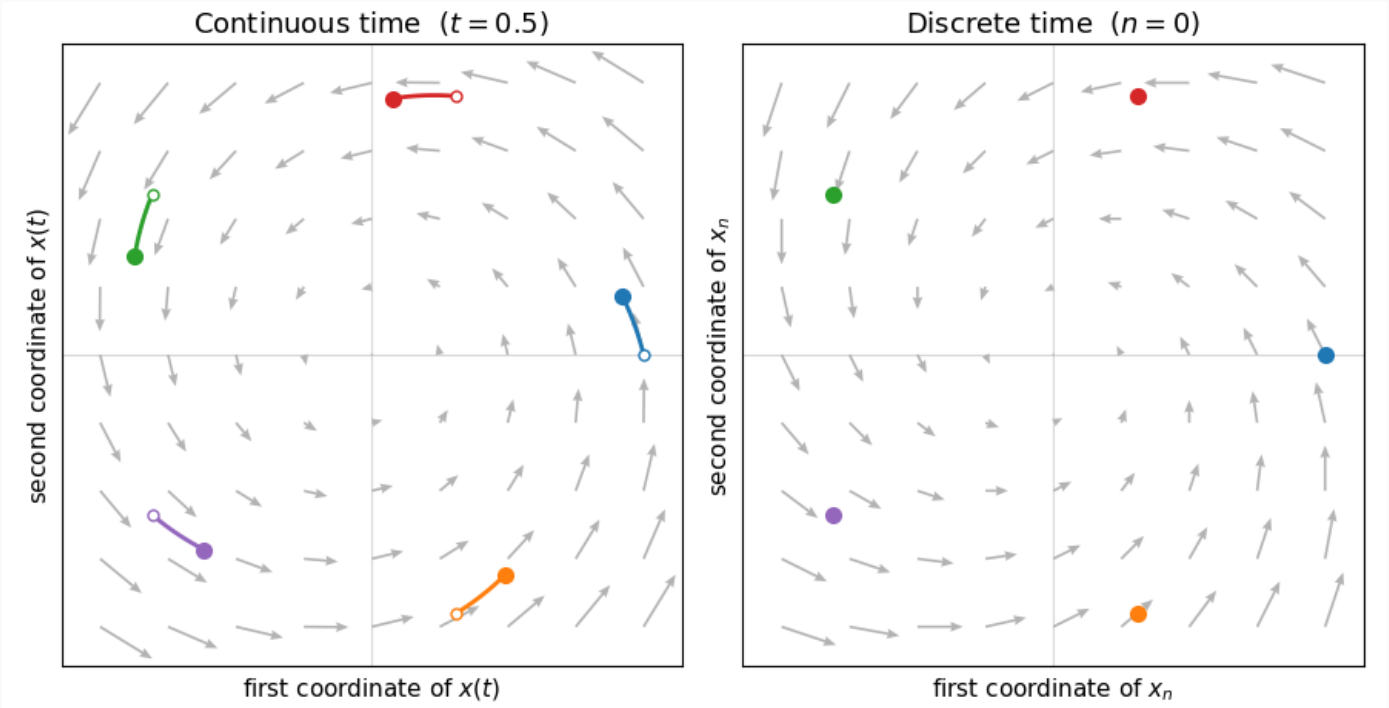


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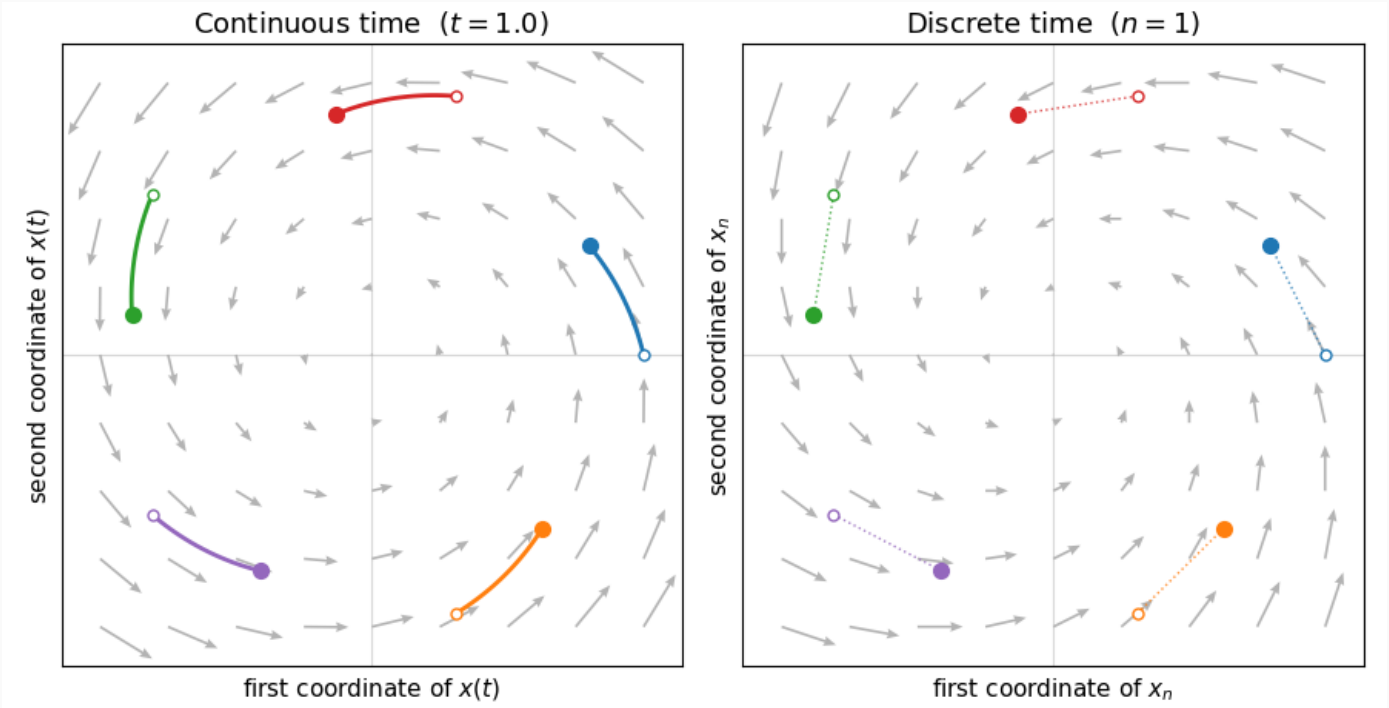


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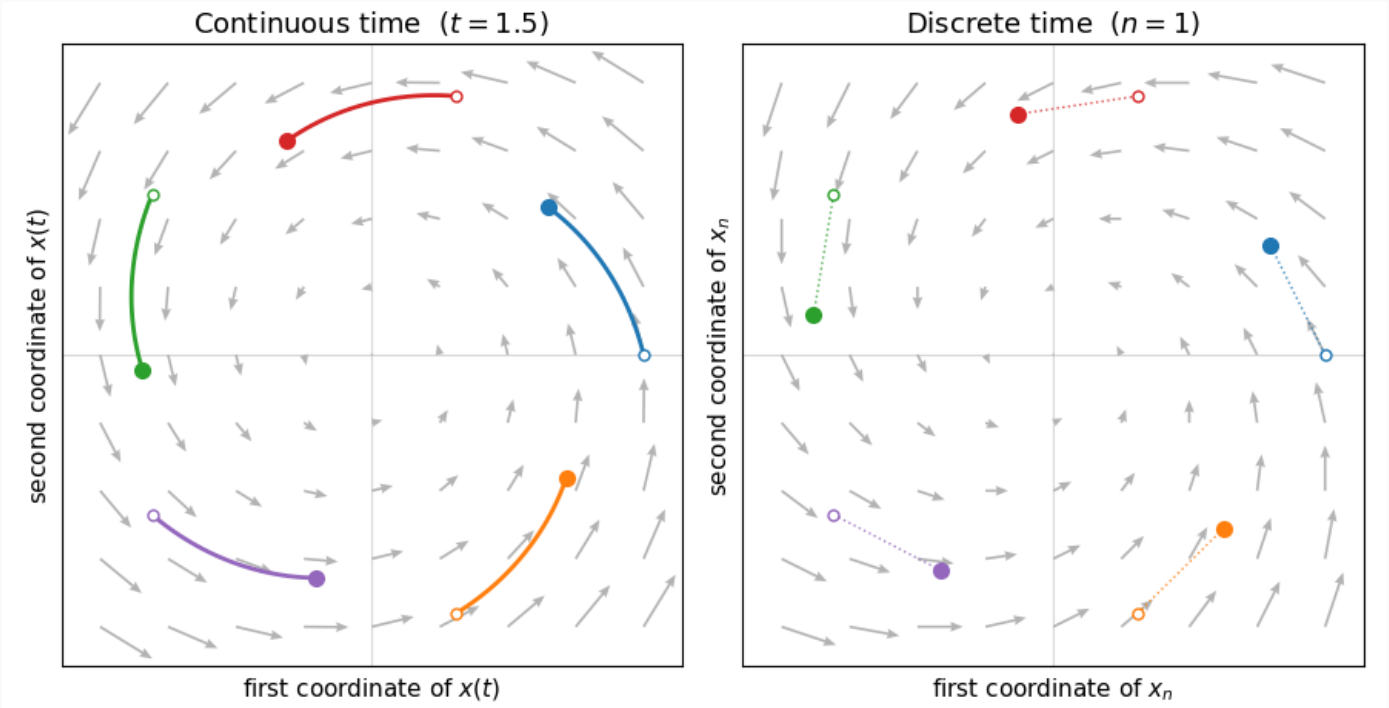


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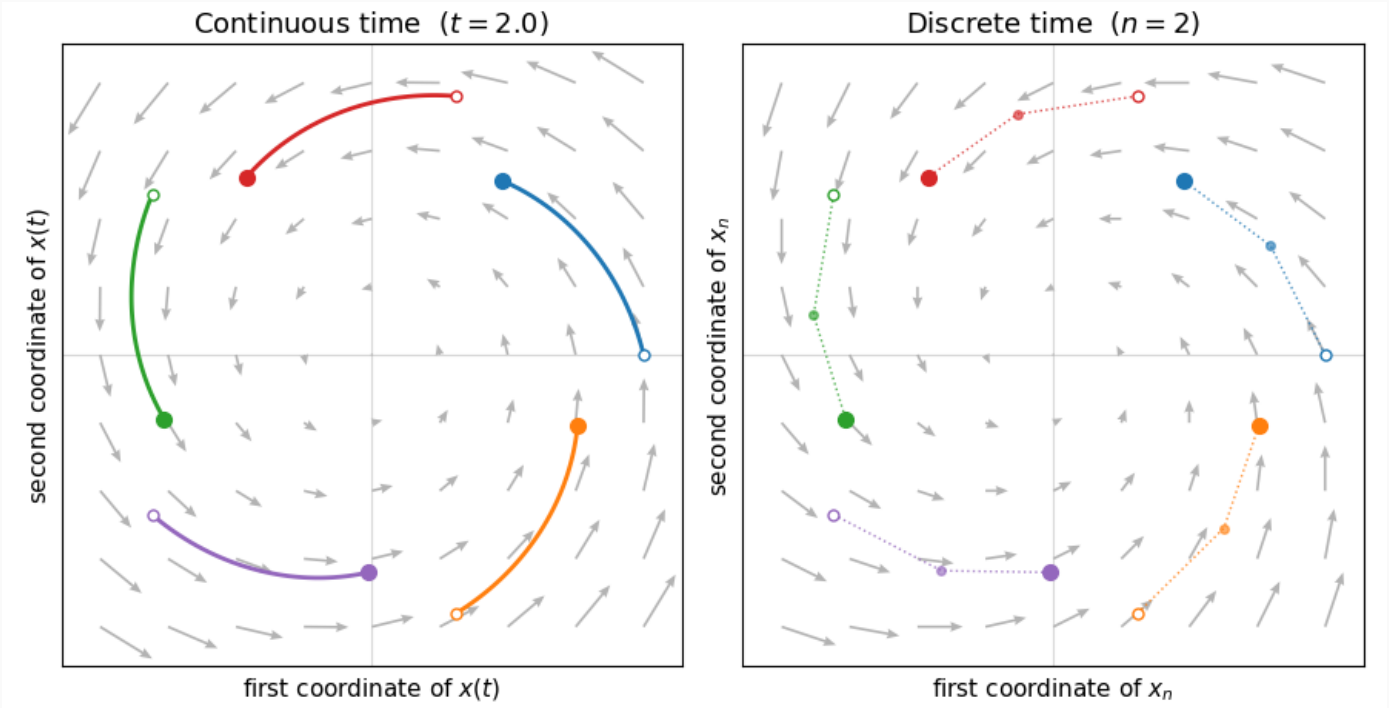


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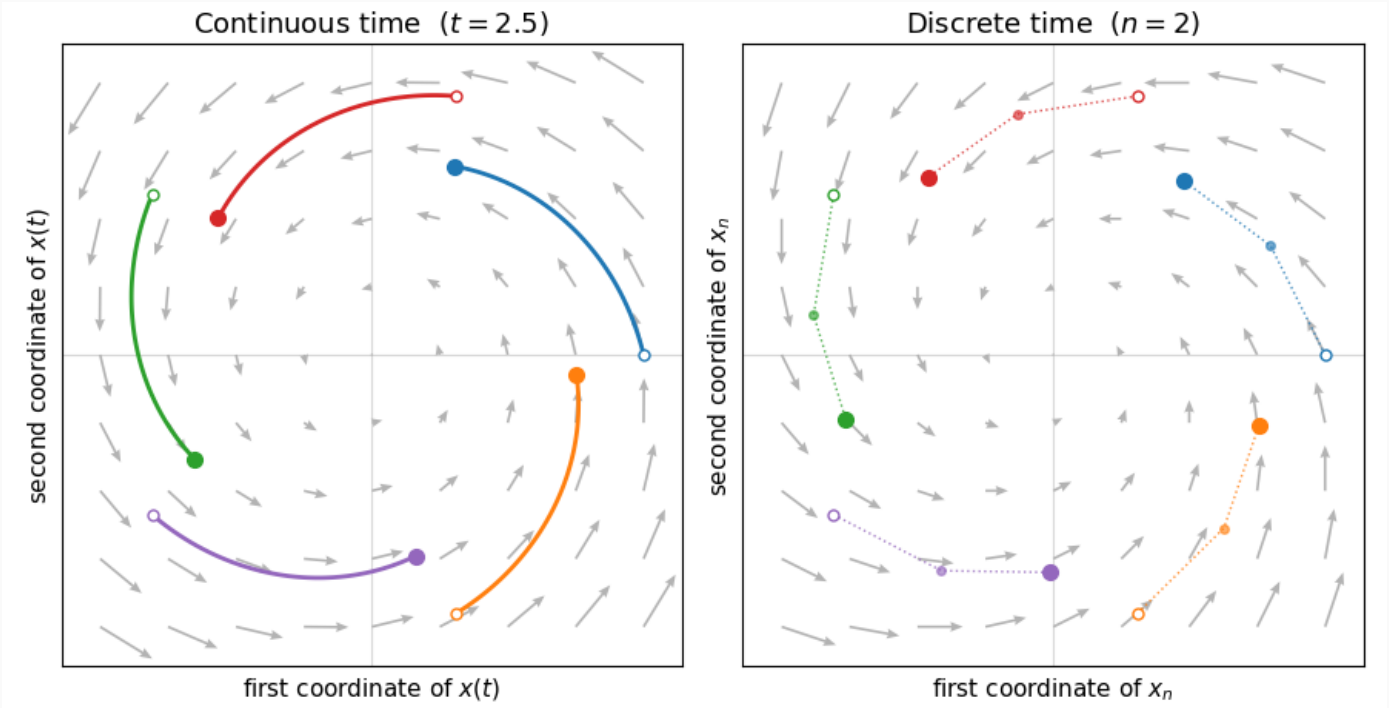


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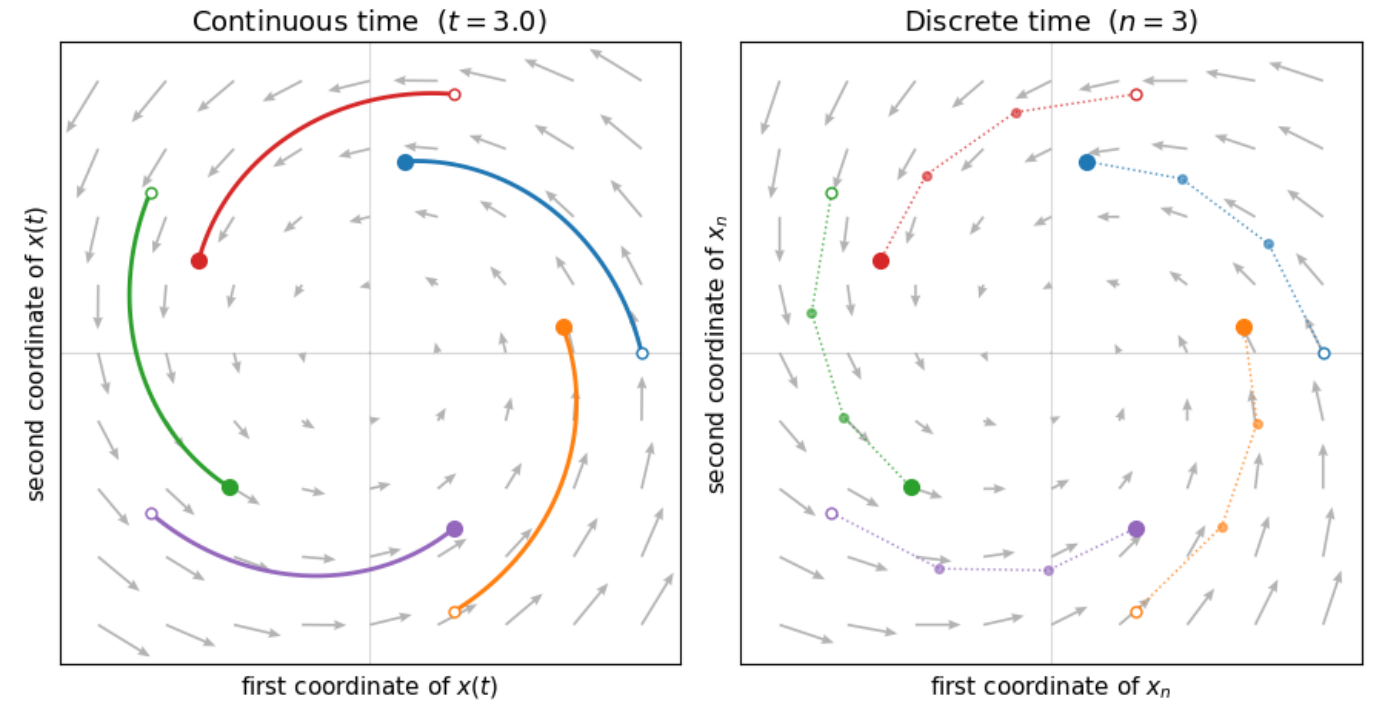


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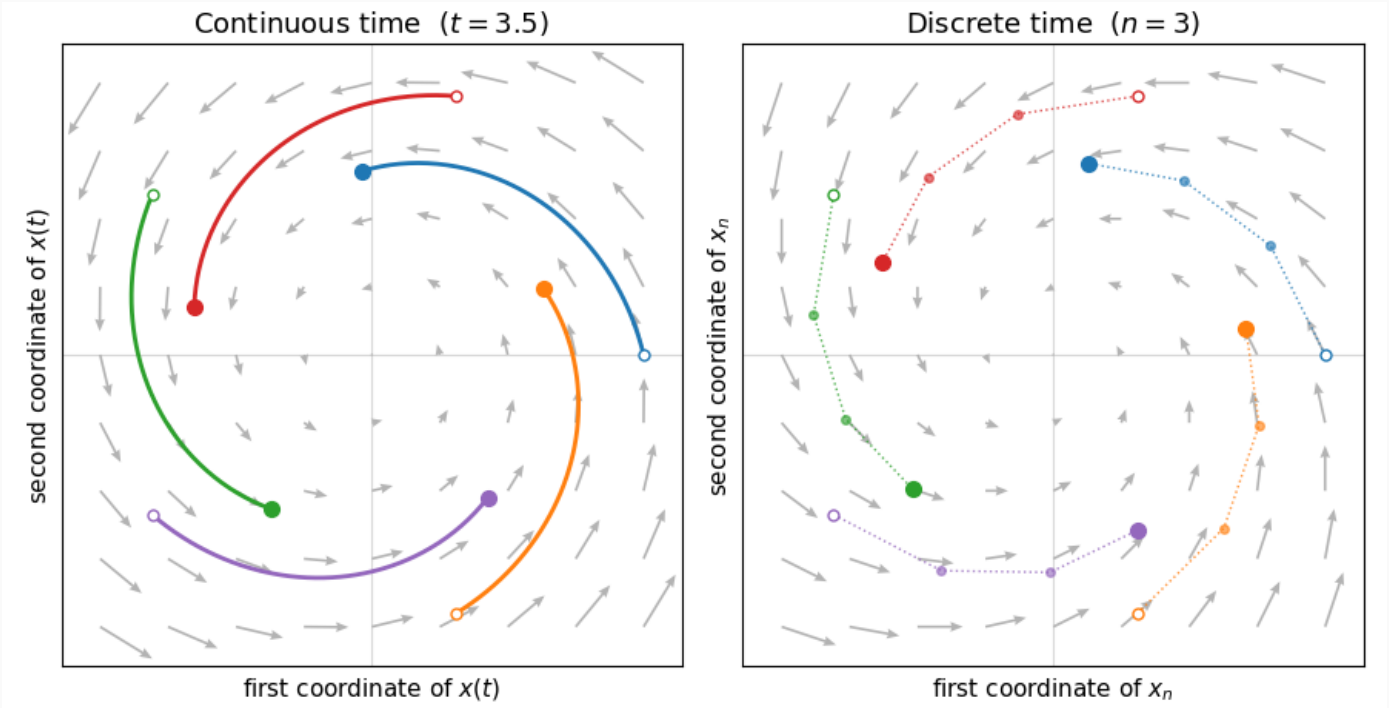


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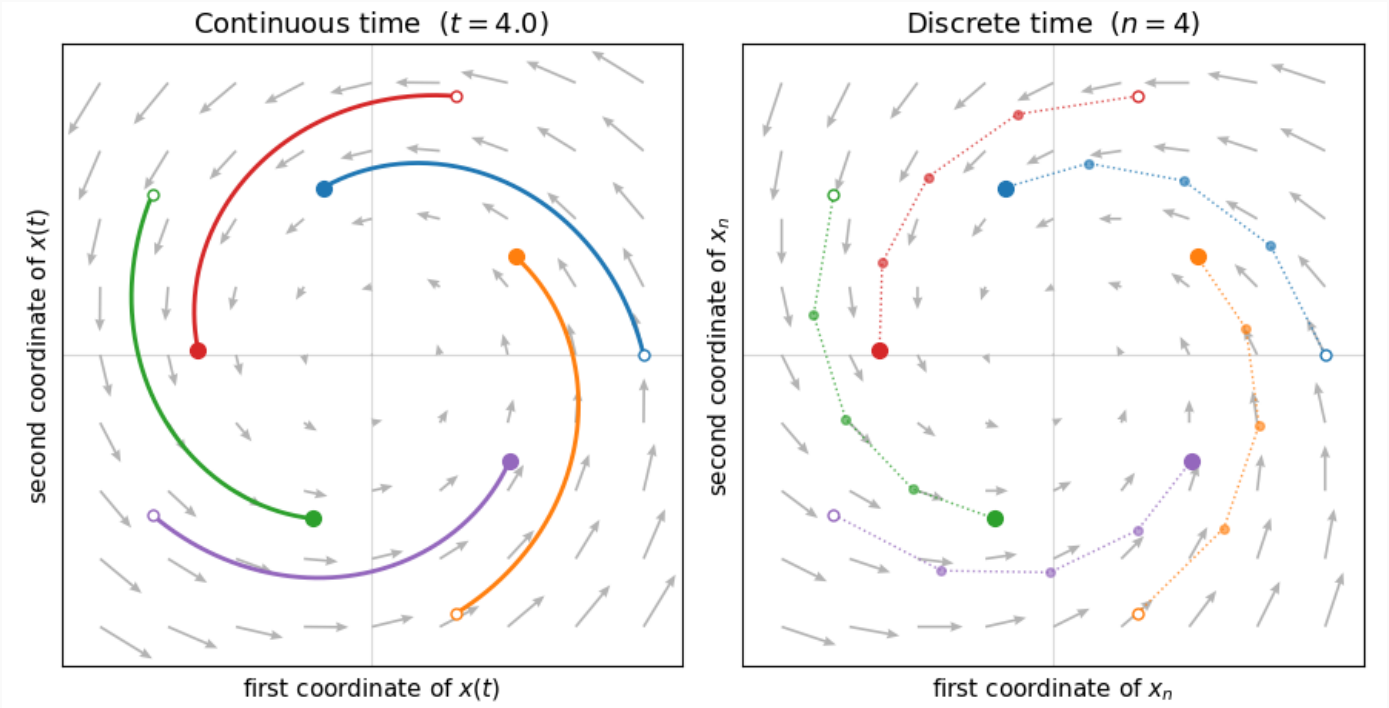


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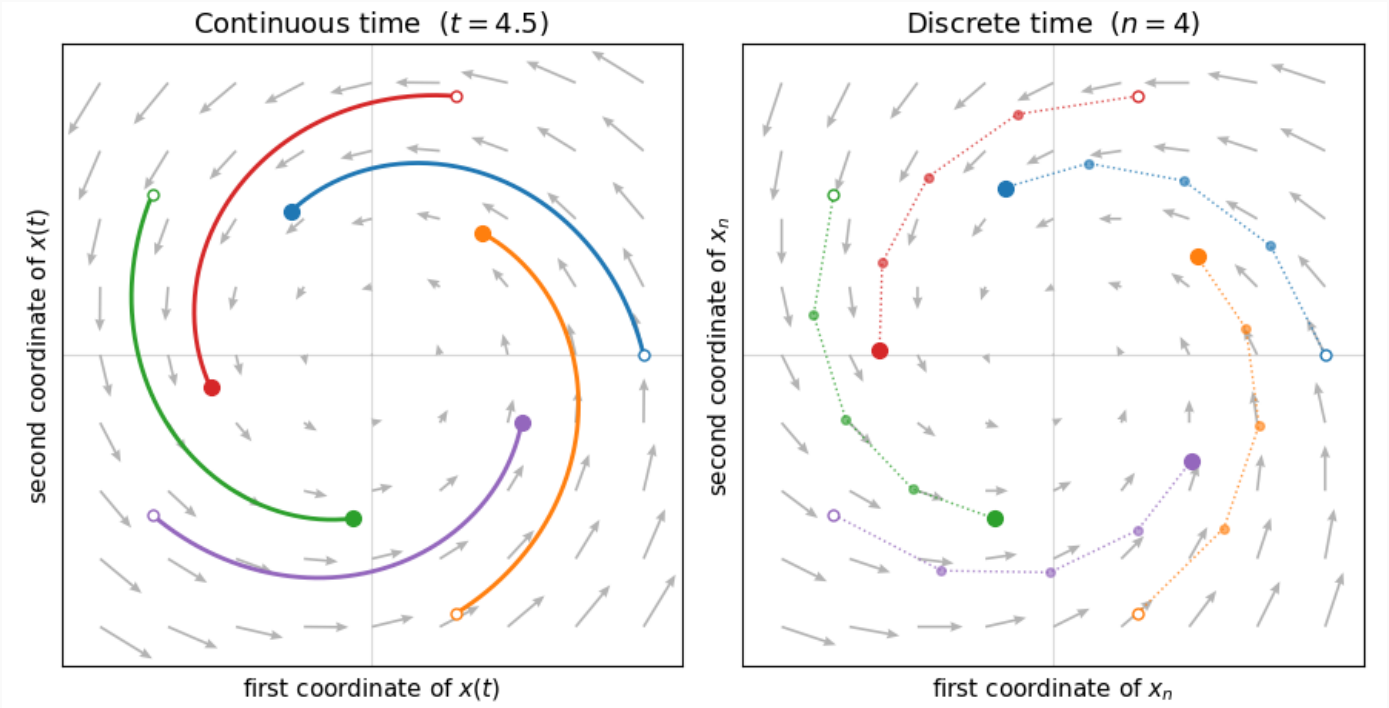


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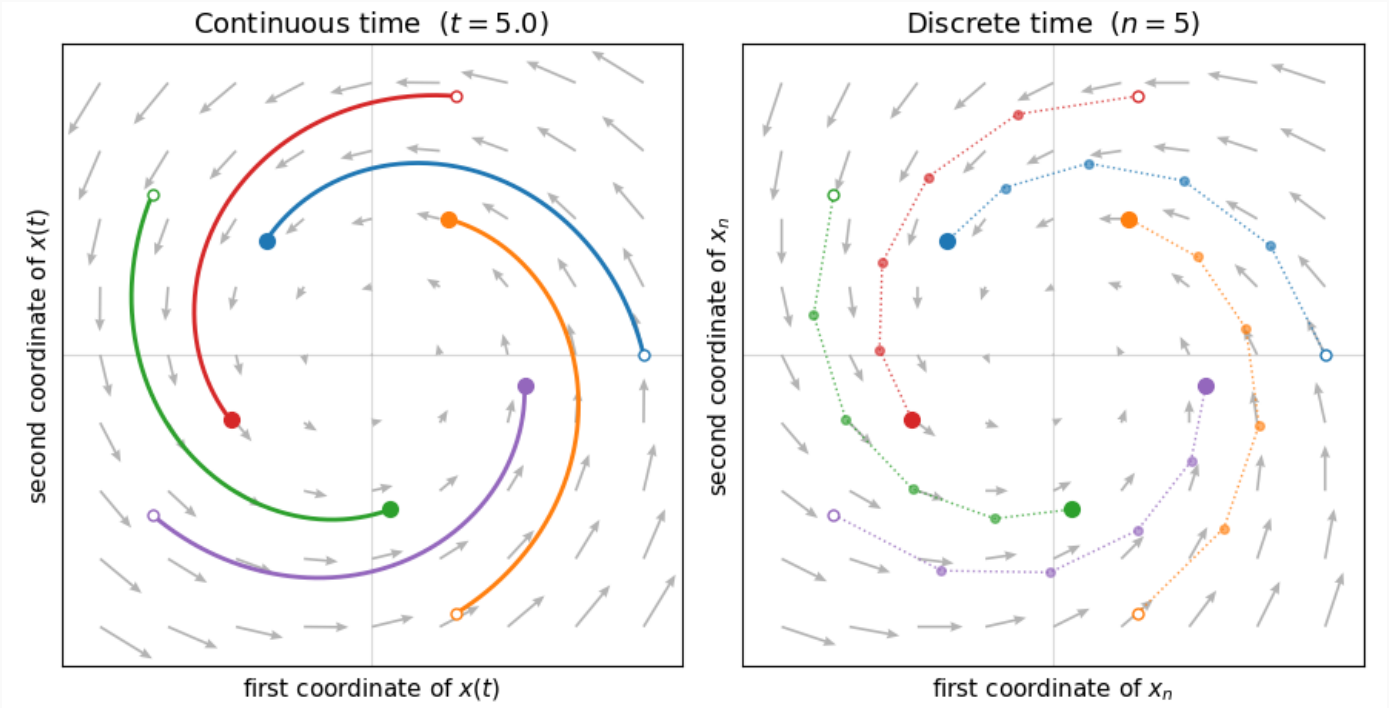


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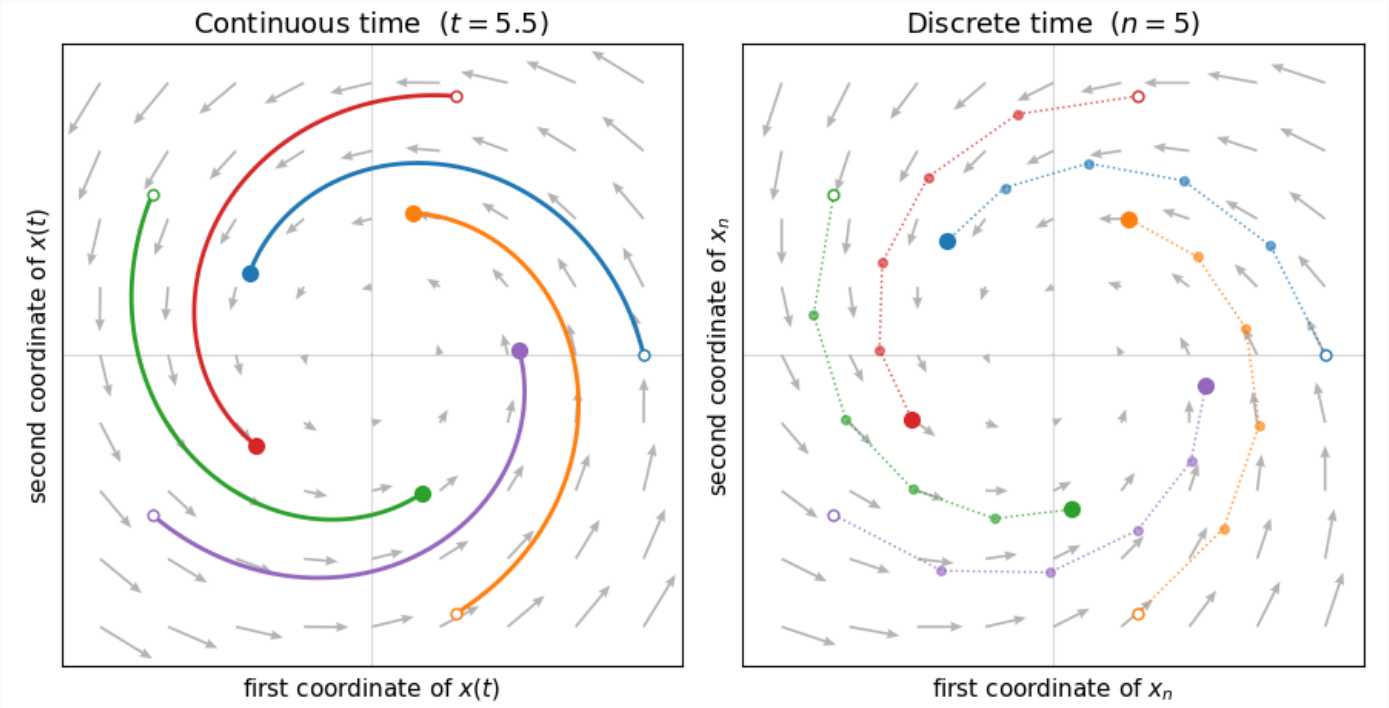


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Below, each coordinate obeys $a_n = 1.6\,a_{n-1} - 0.8\,a_{n-2}$, which gives **spirals**.

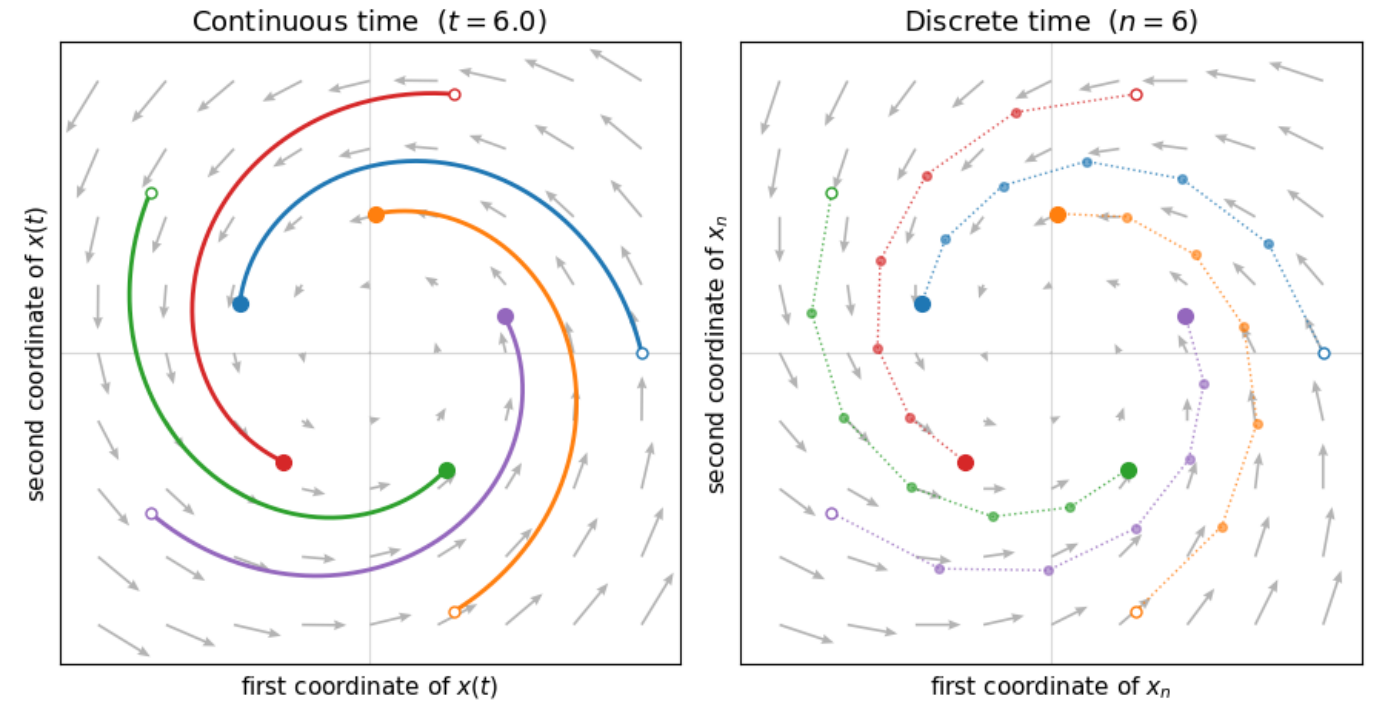


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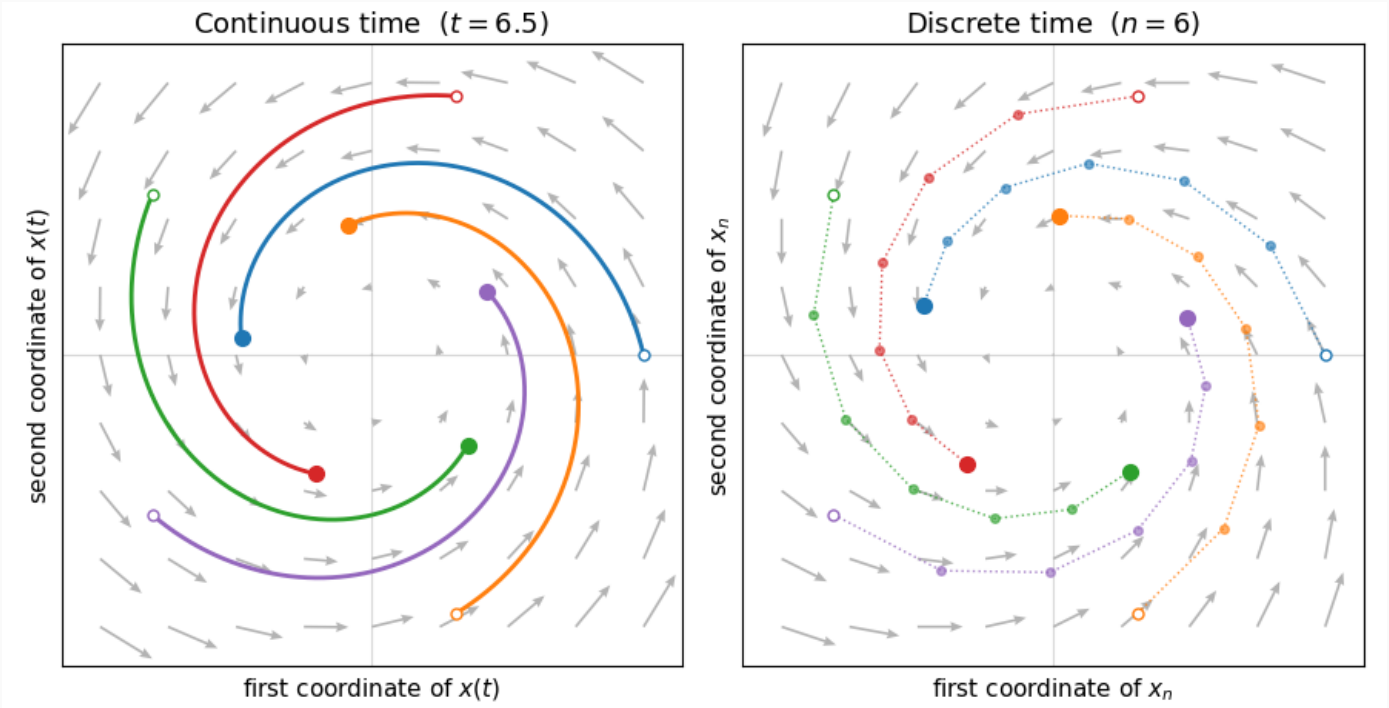


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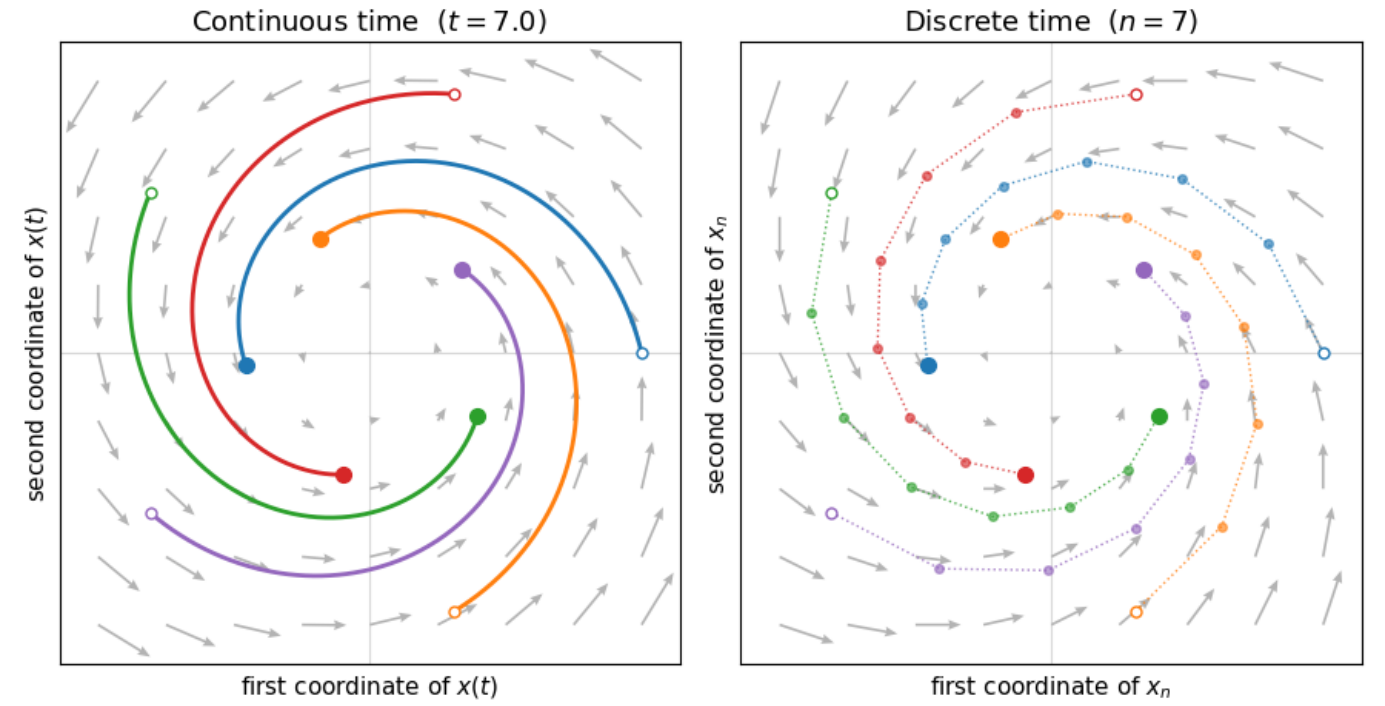


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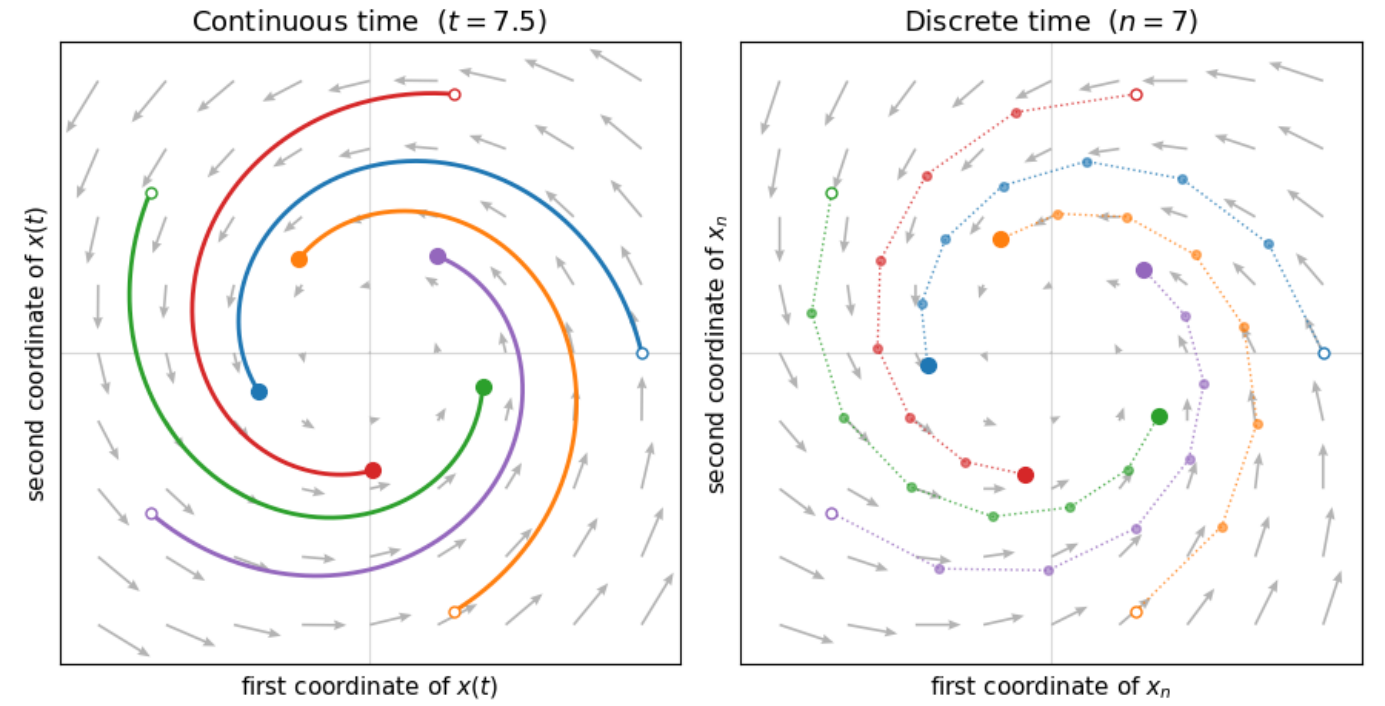


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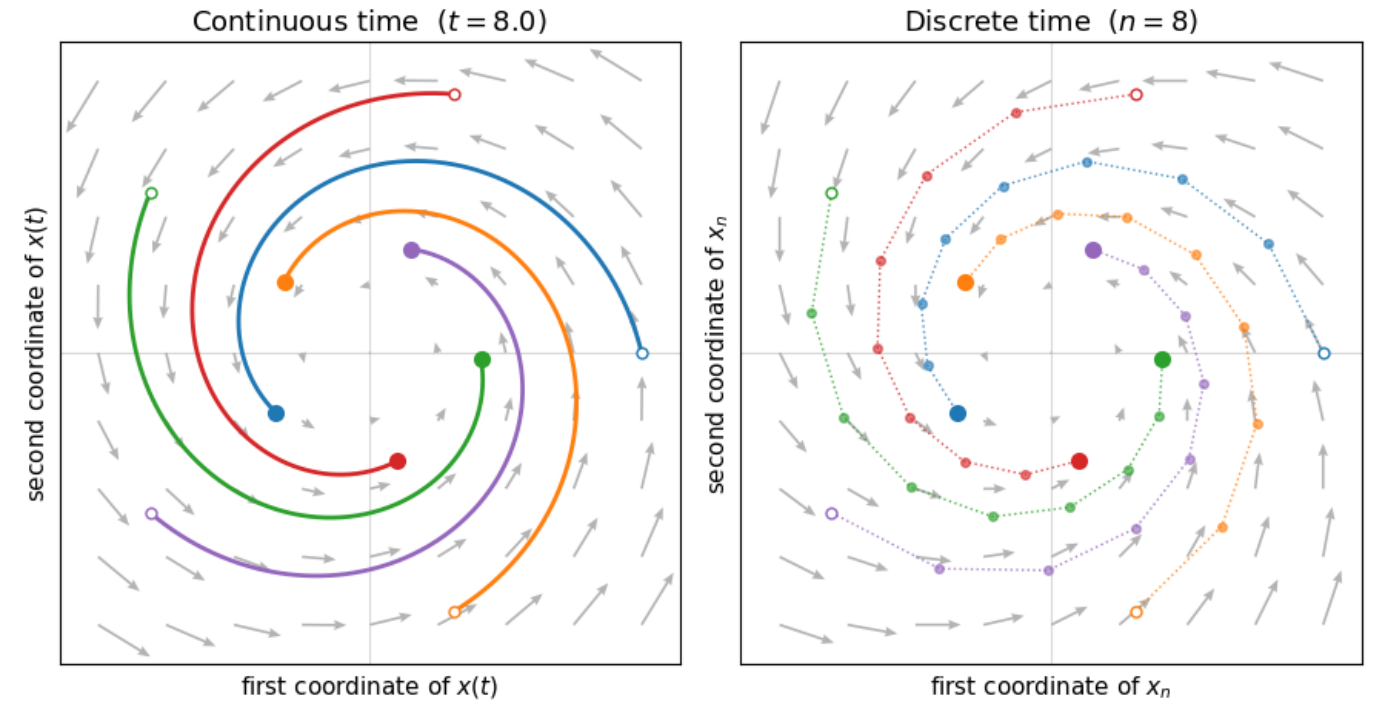


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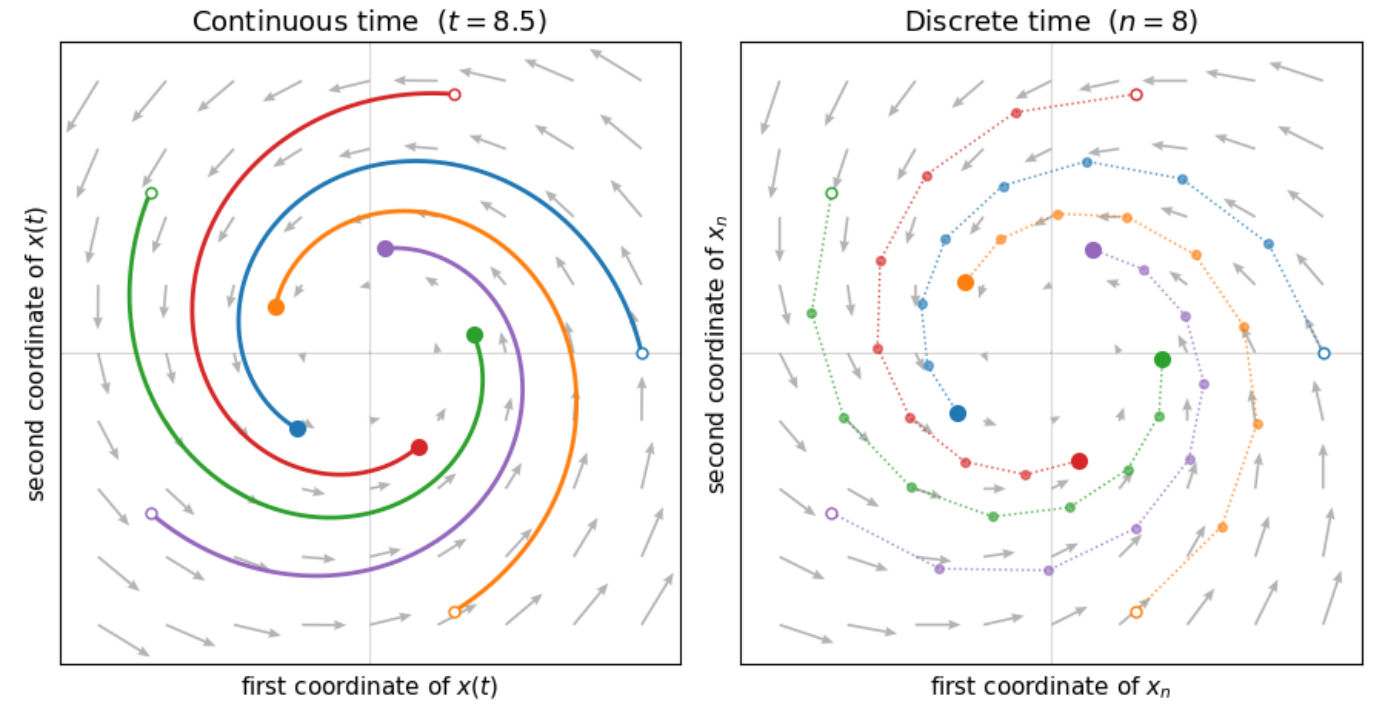


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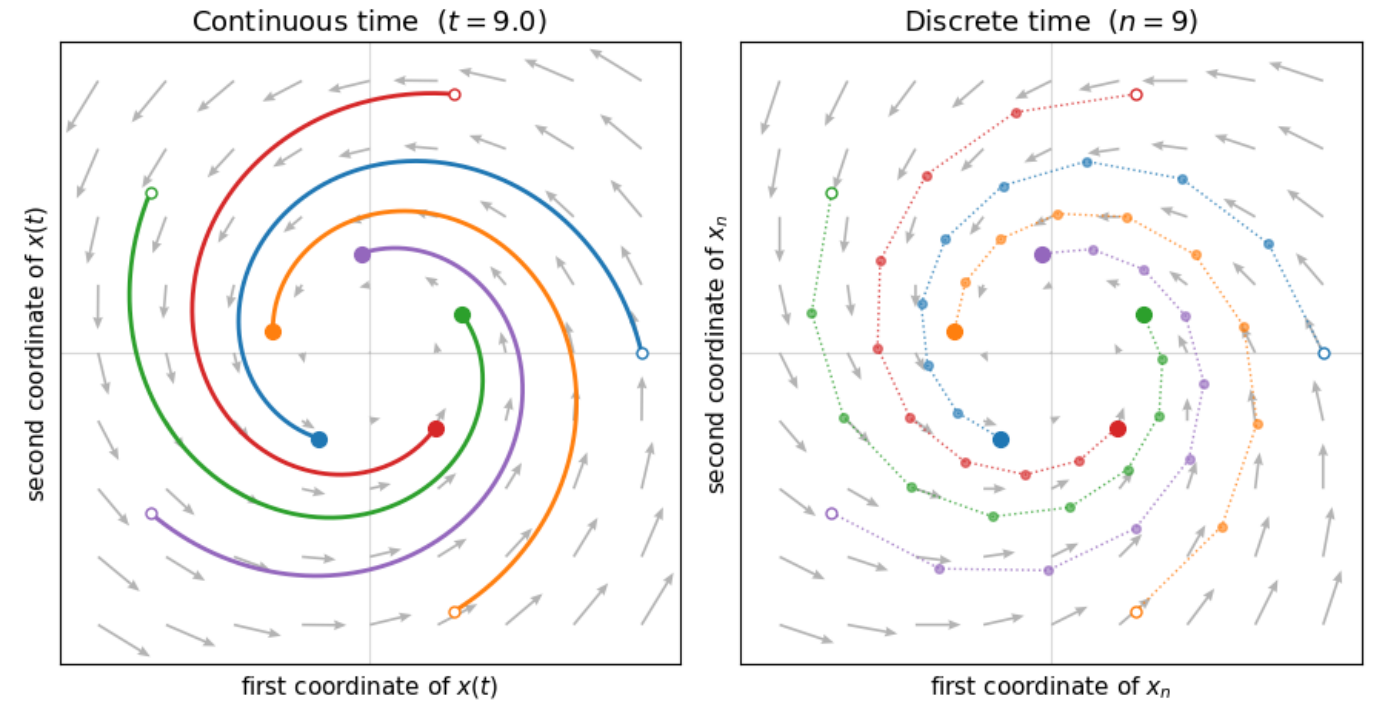


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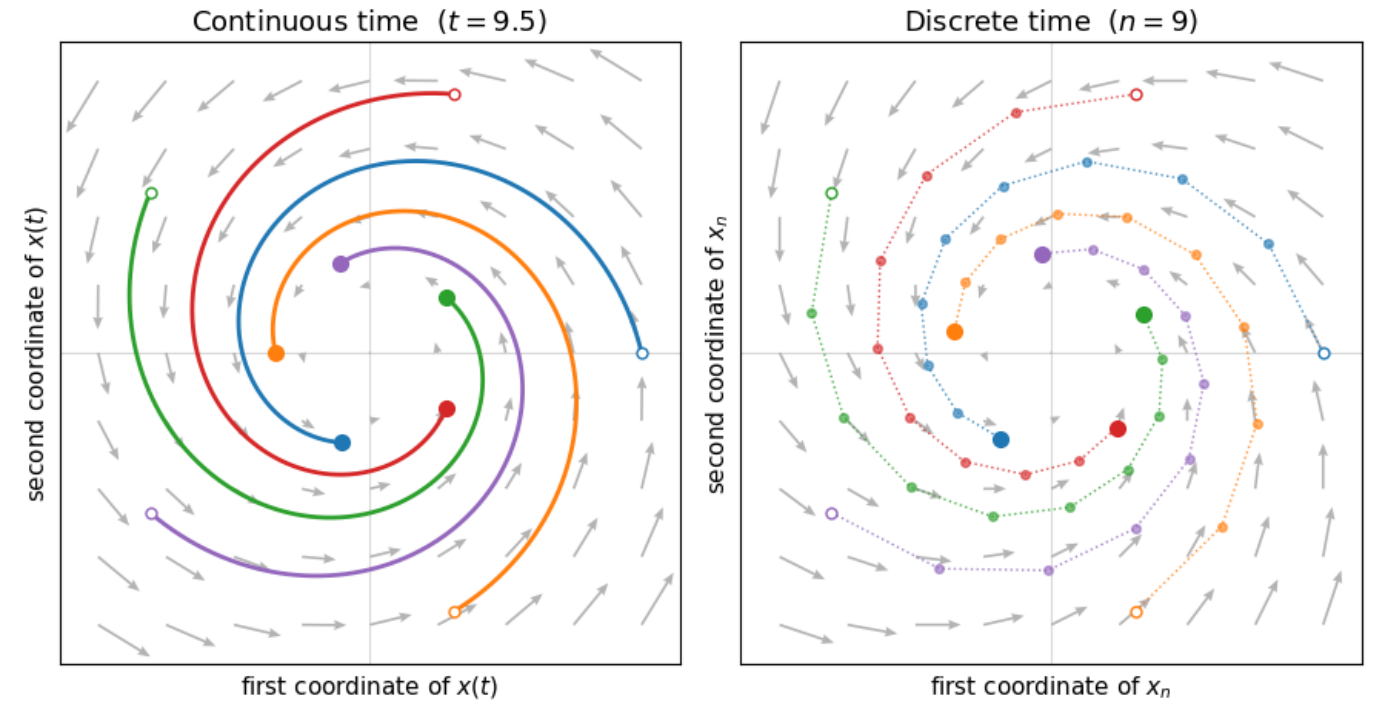


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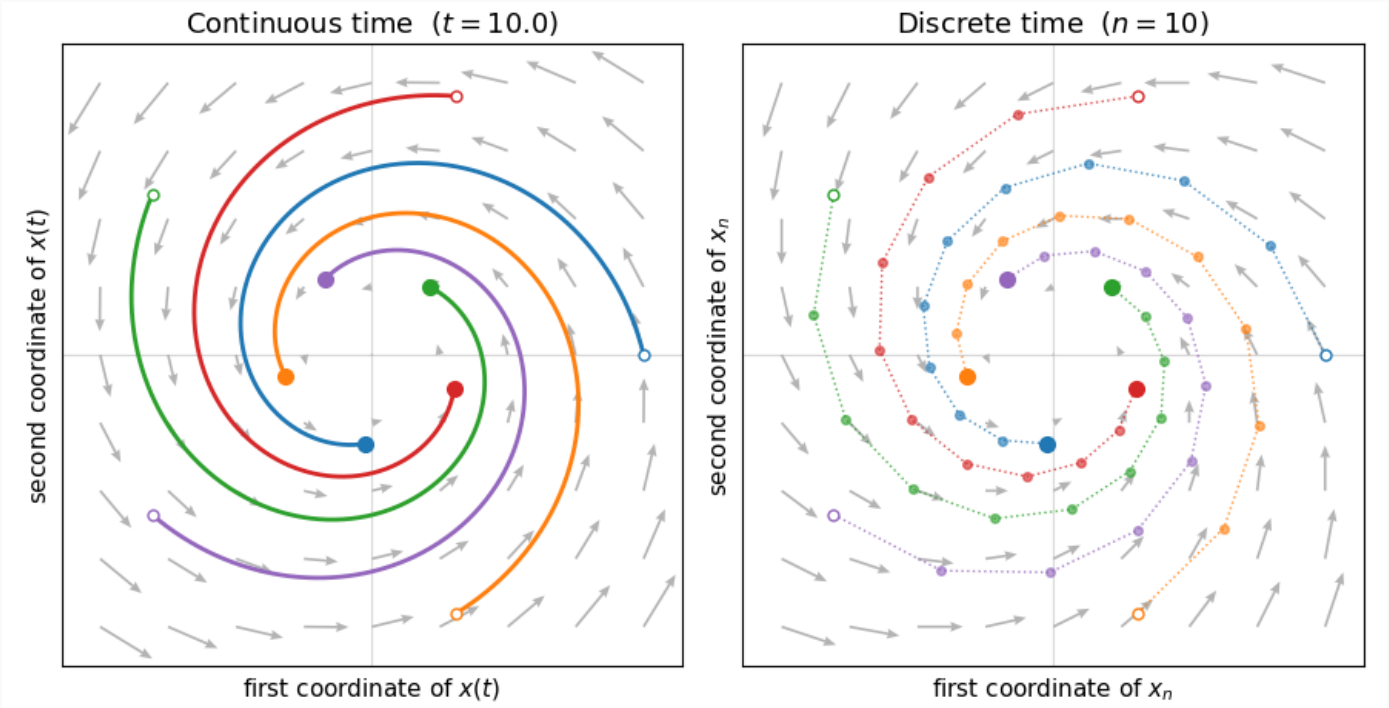


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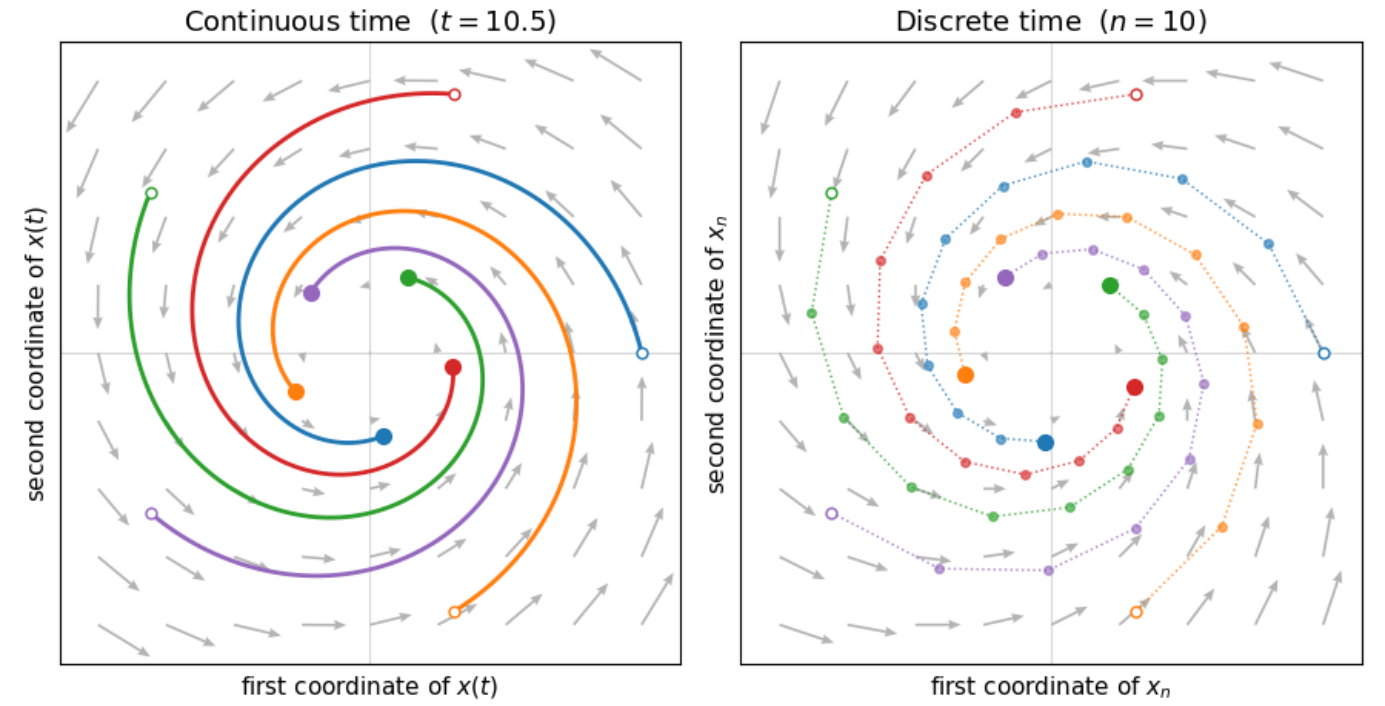


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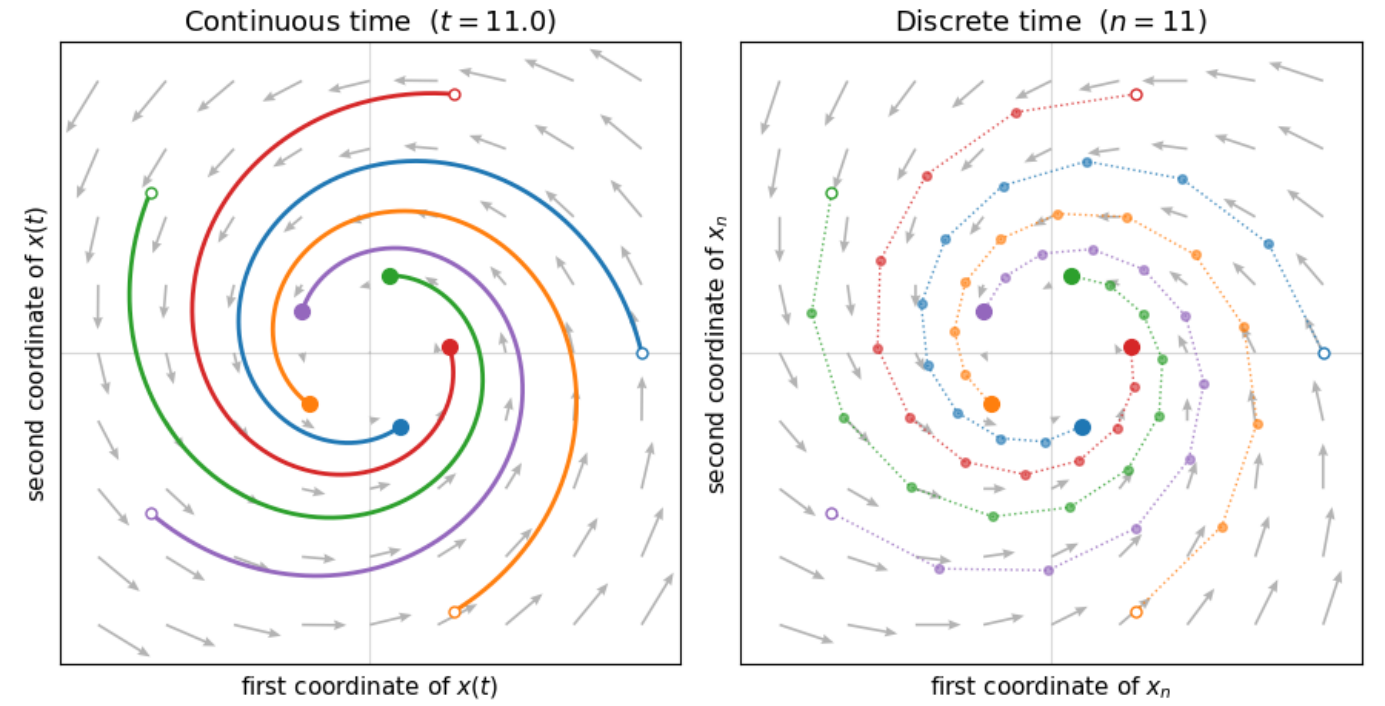


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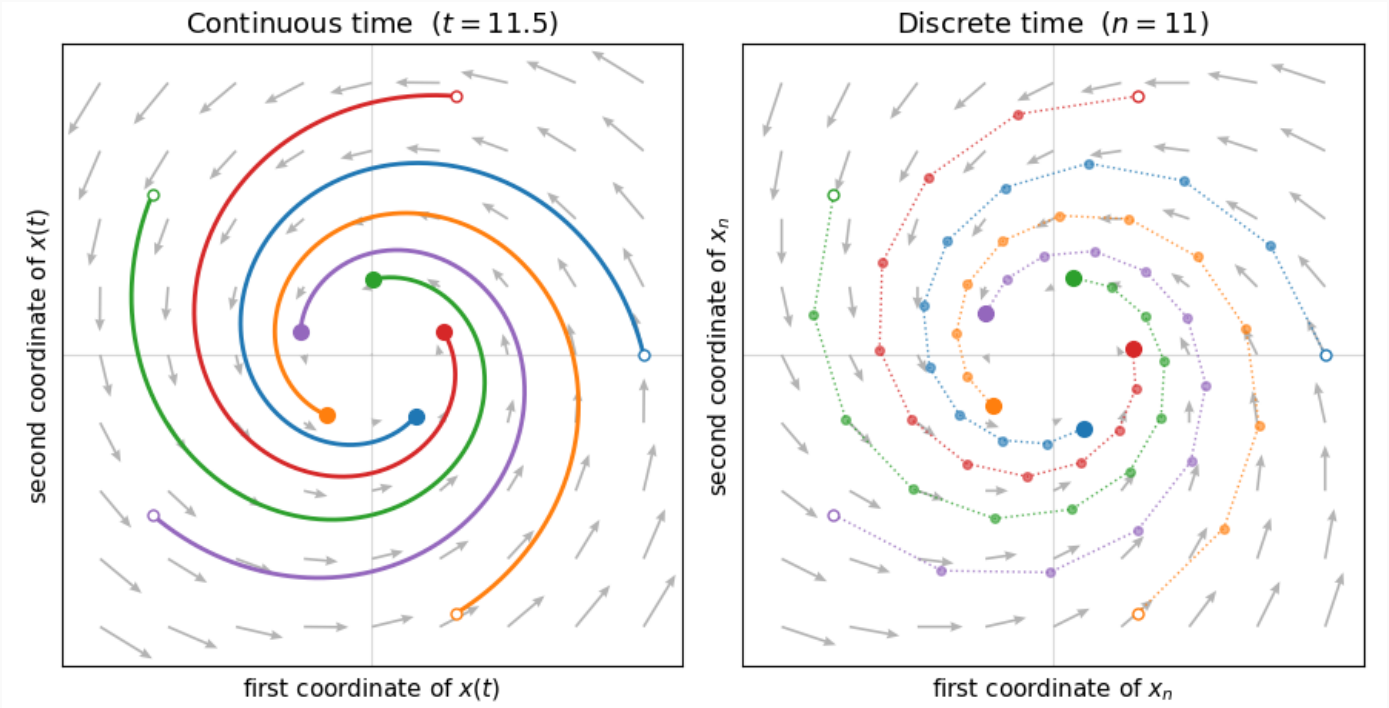


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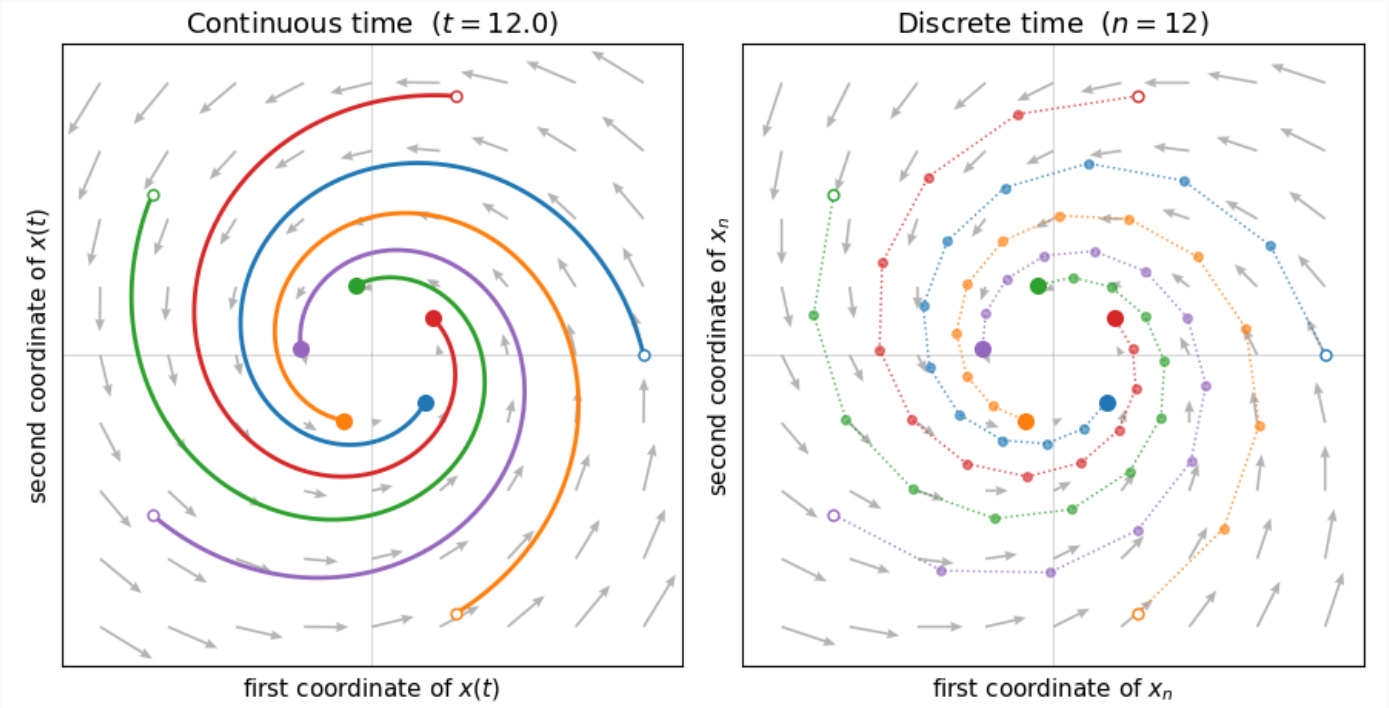


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Group exercises

Random group assignments

Ashley, Isabelle: 7
Bailey, Tanna: 11
Blaisdell, Olyver: 12
Brown, Ashton: 4
Brown, Jonathan: 5
Cardenas, Dominic: 3
Cutler, George: 12
Fabik, Roland: 9
Fellows, Audrey: 2
Fitzgerald, Ryan: 10
Fuller, Ryan: 10
Griffen, Clara: 8
Grutsch, Tanner: 7
Hager, Nathan: 9
Horner, Jayden: 5
Johnson, Jaret: 11
Kintzel, Samantha: 1
Lameres, Alexis: 6
Lee, Holden: 1
Mayala, Reece: 8
McLean, Connor: 10
Mcbride, Daniel: 7
Miller, Maverick: 6
Moss, Emily: 13
Mousseau, Matthew: 2
Neuman, Evan: 11
Putnam, John: 8
Ramos, Vydel: 4
Rathbun, Jedidiah: 5
Roberts, Matthew: 12
Roller, Nathan: 3
Ryan, Otis: 13
Shevtsov, Timothy: 1
Sinclair, Aidan: 9
Stensland, Emma: 6
Thiessen, Logan: 2
Thompson, Jaiden: 3
Vath, Ethan: 4

Group exercises

1. Solve the Tower of Hanoi recursion.
2. Solve each of the following recurrence relations by giving an explicit formula for a_n. For each, please calculate a_9.

Problem	Recurrence relation	Initial condition
1a.	$a_n = 3a_{n-1} - 1$	$a_0 = 10$
1b.	$a_n = a_{n-1} + 5$	$a_0 = 10$
1c.	$a_n = 8a_{n-1} - 15a_{n-2}$	$a_0 = 1,\ a_1 = 4$
1d.	$a_n = 4a_{n-1} - 4a_{n-2}$	$a_0 = 5,\ a_1 = 1$

3. Some so-called intelligence tests often include problems in which a series of numbers is presented and the subject is required to find the next term of the sequence. For example, the sequence might begin 1,2,4,8. No doubt the examiner is looking for 16 as the next term.

   Show how to "outsmart" the intelligence test by finding a polynomial expression (of degree 3) for a_n such that $a_0 = 1, a_1 = 2, a_2 = 4, a_3 = 8$, but $a_4 = 15$.

Solution to group exercise #1

Problem. Solve the Tower of Hanoi recursion.

Remark. Recall that the Tower of Hanoi recursion is

$$m_k = 2m_{k-1} + 1, \tag{1}$$

where m_k is the minimum number of moves needed to transfer a tower of k disks from one pole to another.

Solution. By Prop 23.1, we know the solution satisfies

$$m_k = c_1 2^k + c_2 \tag{2}$$

for some constants c_1, c_2. From the structure of the Tower of Hanoi problem, we have the initial condition $m_1 = 1$. Propagating this through the recursion in Eq. (1), we obtain $m_2 = 3$. By substitution into the form of the solution in Eq. (2), we obtain

$$1 = 2c_1 + c_2$$

$$3 = 4c_1 + c_2$$

Solving this system of equations yields $c_1 = 1, c_2 = -1$, and so

$$m_k = 2^k - 1.$$

Solution to group exercise #2: Overview

Group exercise #1 asks us to deduce explicit formulas for a_n from recurrence relations.

First Order Recurrence. A first order recurrence takes the form

$$a_n = sa_{n-1} + t$$

The explicit solution depends on the context

Context	Solution	Justification
If $s \neq 1$	$a_n = c_1 s^n + c_2$	Prop 23.1
If $s = 1$	$a_n = a_0 + nt$	Prop 23.3

Second Order Recurrence. A second order recurrence takes the form

$$a_n = s_1 a_{n-1} + s_2 a_{n-2}$$

The explicit solution depends on the context. First, we solve the quadratic formula $x^2 - s_1 x - s_2 = 0$ to find roots r_1 and r_2.

Context	Solution	Justification
If $r_1 \neq r_2$	$a_n = c_1 r_1^n + c_2 r_2^n$	Thm 23.5
If $r_1 = r_2 := r$	$a_n = c_1 r^n + c_2 nr^n$	Thm 23.9

Remark. Whenever the solution involves unknown constants (denoted c_1 and c_2), you must solve for them. You do this by plugging in the first two values for the sequence (a_0 and a_1) along with other known values (some subset of $\{s^n, t, r_1^n, r_2^n, r^n, nr^n\}$) to form a system of two equations with two unknowns. Then solve the system (e.g. using substitution).

Solution to group exercise #2a

Problem. Solve the recurrence relation $a_n = 3a_{n-1} - 1$ with initial condition $a_0 = 10$ to give an explicit formula for a_n.

Solution. By Prop 23.1, the solution takes the form

$$a_n = c_1 s^n + c_2$$
$$\implies a_n = c_1 \cdot 3^n + c_2$$

Substituting in the provided initial condition $a_0 = 10$ and the next term of the sequence $a_1 = 29$, we obtain the system of equations

$$10 = c_1 + c_2$$
$$29 = 3c_1 + c_2$$

Solving this system yields $c_1 = 9.5$ and $c_2 = 0.5$. Hence, the final solution is

$$a_n = 9.5 \cdot 3^n + 0.5$$

Solution to group exercise #2b

Problem. Solve the recurrence relation $a_n = a_{n-1} + 5$ with initial condition $a_0 = 10$ to give an explicit formula for a_n.

Solution. By Prop 23.2, the solution takes the form

$$a_n = a_0 + nt$$

Substituting in the provided initial condition $a_0 = 10$ and the given $t = 5$, we obtain the final solution

$$a_n = 10 + 5n$$

Solution to group exercise #2c

Problem. Solve the recurrence relation $a_n = 8a_{n-1} - 15a_{n-2}$ with initial conditions $a_0 = 1$ and $a_1 = 4$ to give an explicit formula for a_n.

Solution. First, we solve the quadratic formula

$$\begin{aligned} & x^2 - s_1 x - s_2 = 0 \\ \implies & x^2 - 8x + 15 = 0 \\ \implies & (x-5)(x-3) = 0 \end{aligned}$$

Hence the roots are $r_1 = 5$ and $r_2 = 3$. Since these roots are distinct, we apply Thm 23.5, which tell us that the solution takes the form

$$\begin{aligned} & a_n = c_1 r_1^n + c_2 r_2^n \\ \implies & a_n = c_1 5^n + c_2 3^n \end{aligned}$$

Substituting in the provided initial conditions $a_0 = 1$ and $a_1 = 4$, we obtain the system of equations

$$\begin{aligned} 1 &= c_1 + c_2 \\ 4 &= 5c_1 + 3c_2 \end{aligned}$$

Solving this system yields $c_1 = \frac{1}{2}$ and $c_2 = \frac{1}{2}$. Hence, the final solution is

$$a_n = \frac{1}{2} \cdot 5^n + \frac{1}{2} \cdot 3^n$$

Solution to group exercise #2d

Problem. Solve the recurrence relation $a_n = 4a_{n-1} - 4a_{n-2}$ with initial conditions $a_0 = 5$ and $a_1 = 1$ to give an explicit formula for a_n.

Solution. First, we solve the quadratic formula

$$\begin{aligned} & x^2 - s_1 x - s_2 = 0 \\ \implies & x^2 - 4x + 4 = 0 \\ \implies & (x-2)(x-2) = 0 \end{aligned}$$

Hence the roots are $r_1 = 2$ and $r_2 = 2$. Since these roots are identical, we apply Thm 23.9, which tell us that the solution takes the form

$$\begin{aligned} & a_n = c_1 r^n + c_2 n r^n \\ \implies & a_n = c_1 2^n + c_2 n 2^n \end{aligned}$$

Substituting in the provided initial conditions $a_0 = 5$ and $a_1 = 1$, we obtain the equations

$$\begin{aligned} 5 &= c_1 \\ 1 &= 2c_1 + 2c_2 \end{aligned}$$

Solving these equations yields $c_1 = 5$ and $c_2 = -4.5$. Hence, the final solution is

$$a_n = 5 \cdot 2^n - 4.5 \cdot n 2^n$$

Solution to group exercise #3

Problem. [...] Show how to "outsmart" the intelligence test by finding a polynomial expression (of degree 3) for a_n such that $a_0 = 1, a_1 = 2$, $a_2 = 4, a_3 = 8$, but $a_4 = 15$.

Solution. We apply Scheinerman Theorem 23.17.

a :	1	2	4	8	15
Δa :	1	2	4	7	
$\Delta^2 a$:	1	2	3		
$\Delta^3 a$:	1	1			
$\Delta^4 a$:	0				

Hence $k = 3$ and

$$a_n = a_0\binom{n}{0} + \Delta a_0\binom{n}{1} + \Delta^2 a_0\binom{n}{2} + \Delta^3 a_0\binom{n}{3}$$

$$= 1\binom{n}{0} + 1\binom{n}{1} + 1\binom{n}{2} + 1\binom{n}{3} \qquad \text{(Substitute coefs from table}$$

$$= [1 \cdot 1] + [1 \cdot n] + \left[1 \cdot \frac{n(n-1)}{2}\right] + \left[1 \cdot \frac{n(n-1)(n-2)}{6}\right] \qquad \text{(Expand binomial coefs.)}$$

$$= \frac{1}{6}n^3 + \frac{5n}{6} + 1 \qquad \text{(Algebra)}$$